

VanController rev 0.2. 12 V vehicle controller: ESP32-S3, 6 fused relays, 6 fused PWM dimmers, CAN, RS485, GPS, accelerometer, 2 water-level inputs.
See DESIGN_SPEC.md for the GPIO map, I2C addresses and design notes.

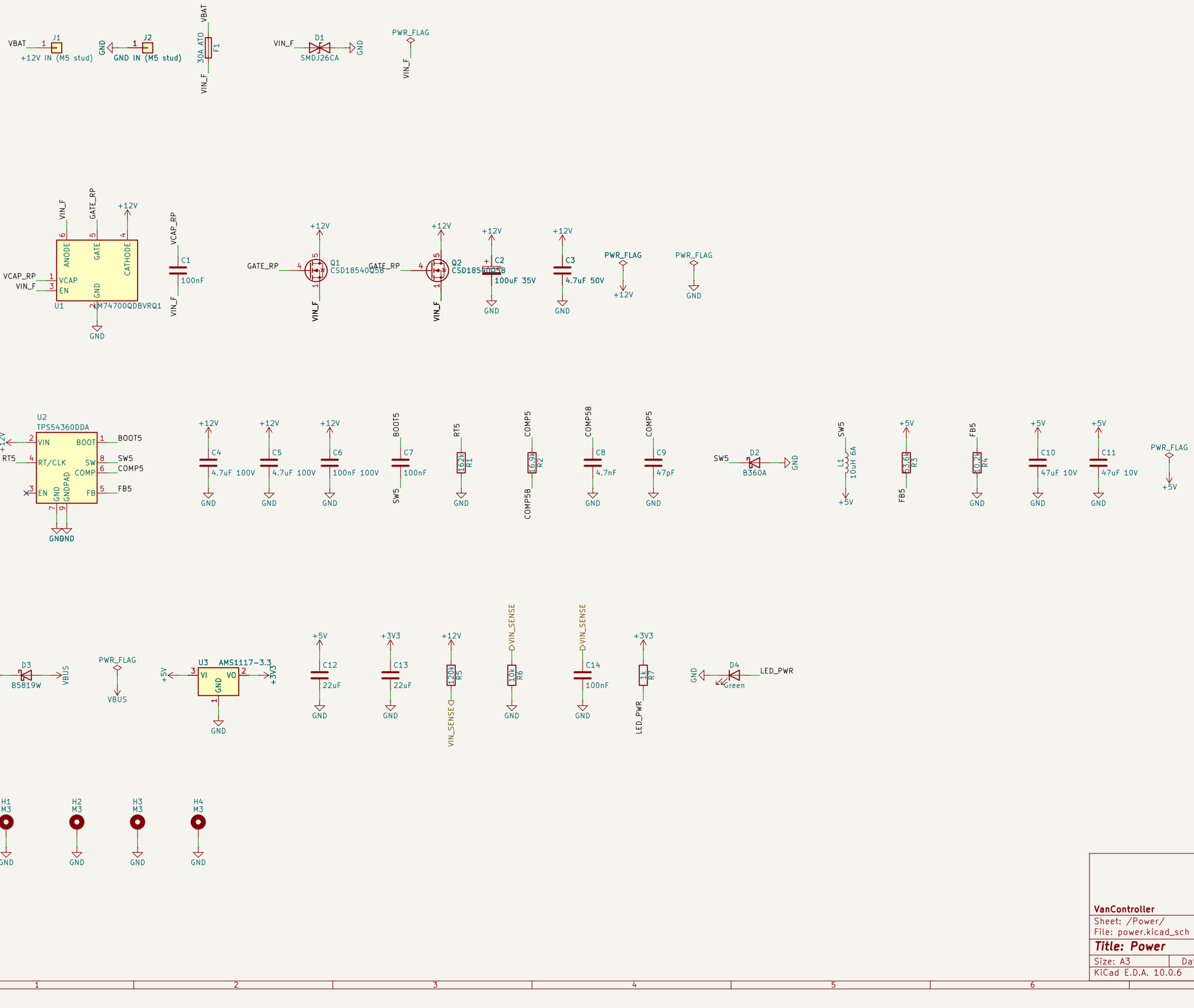
VanController

Sheet: /
File: VanController.kicad_sch

Title: VanController: ESP32-S3 12 V controller

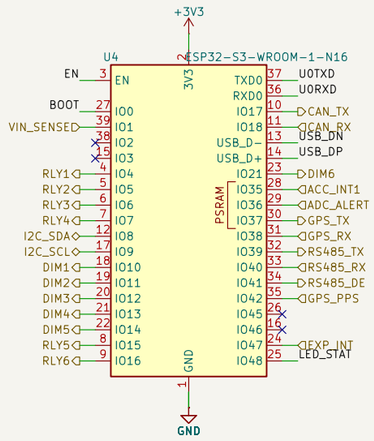
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 1/19

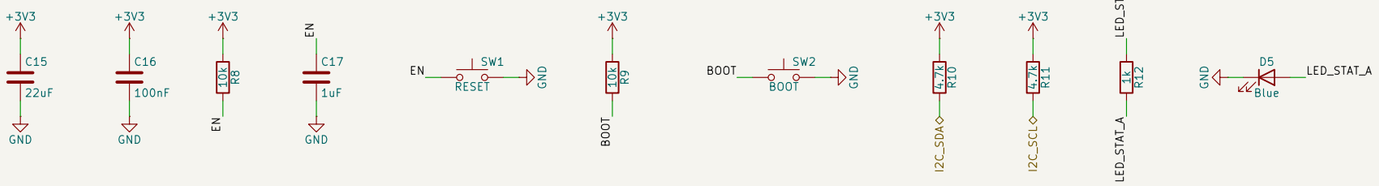
Power: 12 V in -> 30 A ATO main fuse -> bidirectional load-dump TVS -> LM74700 ideal diode with 2 x CSD18540Q5B in parallel (reverse polarity) -> +12V.
TPS54360 buck 12 V -> 5 V (2 A, 600 kHz, values per TI datasheet example; verify with WEBENCH). USB VBUS ORed into +5V via B5819W. AMS1117-3.3 -> +3V3.

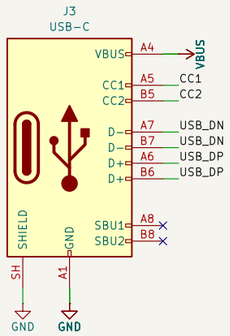


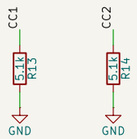
VanController		
Sheet: /Power/		
File: power.kicad_sch		
Title: Power		
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 2/19

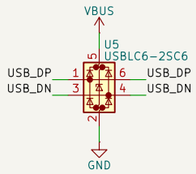
MCU: ESP32-S3-WROOM-1. Strapping pins GPIO0 (BOOT button), 3, 45, 46 left at defaults. Place the module at the board edge with the antenna overhanging or over a copper keep-out on all layers.

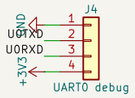






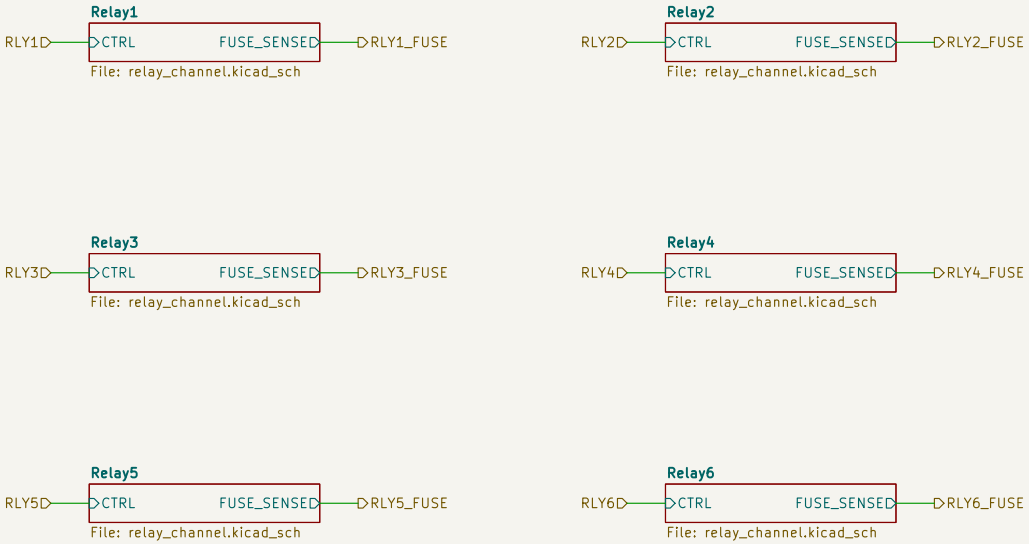






VanController		
Sheet: /MCU/		
File: mcu.kicad_sch		
Title: MCU		
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 3/19

Six identical instances of relay_channel.kicad_sch. Edit that file once to change every channel.



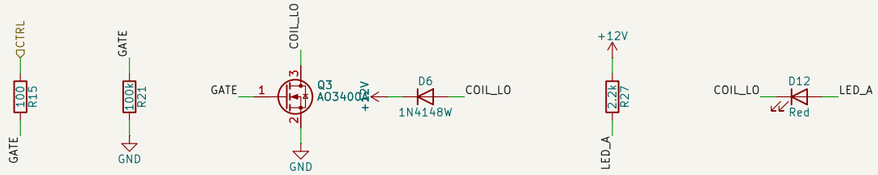
VanController

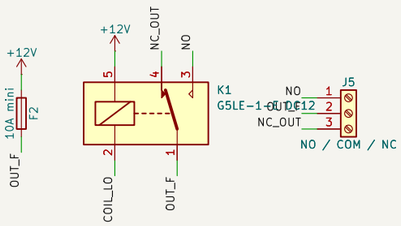
Sheet: /Relays/
File: relays.kicad_sch

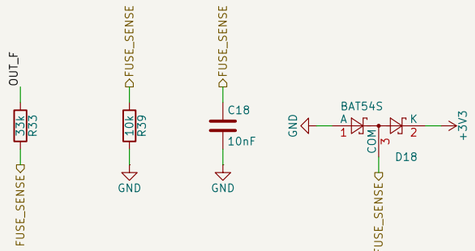
Title: Relays (6 channels)

Size: A4	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 4/19

Relay channel. COM is fed from +12V through the channel fuse, so NO gives switched, fused +12 V. FUSE_SENSE is high (~2.8 V at 12 V) while the fuse is intact; BAT54S clamps transients to +3V3.

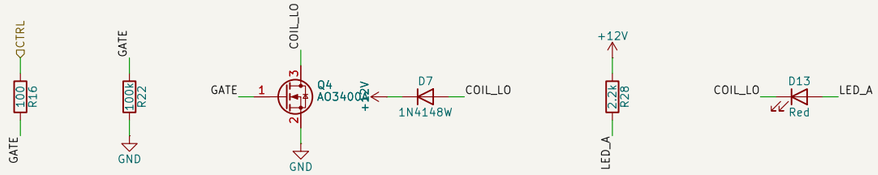


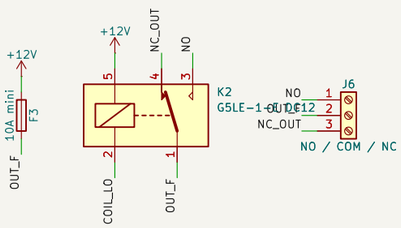


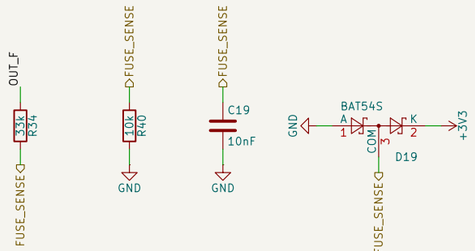


VanController		
Sheet: /Relays/Relay1/		
File: relay_channel.kicad_sch		
Title: Relay channel		
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 5/19

Relay channel. COM is fed from +12V through the channel fuse, so NO gives switched, fused +12 V. FUSE_SENSE is high (~2.8 V at 12 V) while the fuse is intact; BAT54S clamps transients to +3V3.

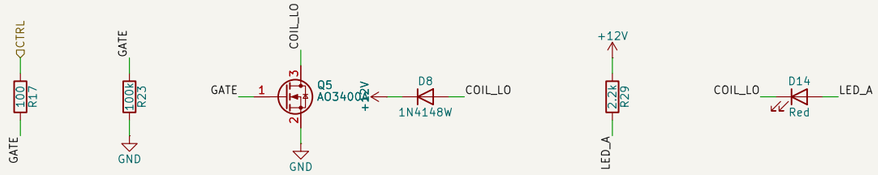


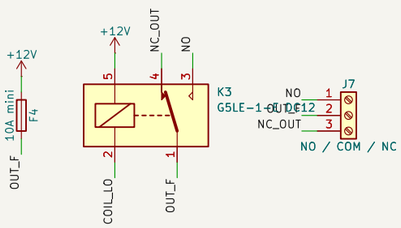


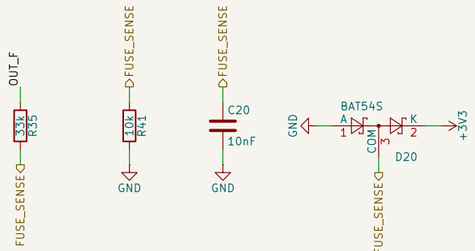


VanController		
Sheet: /Relays/Relay2/		
File: relay_channel.kicad_sch		
Title: Relay channel		
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 6/19

Relay channel. COM is fed from +12V through the channel fuse, so NO gives switched, fused +12 V. FUSE_SENSE is high (~2.8 V at 12 V) while the fuse is intact; BAT54S clamps transients to +3V3.

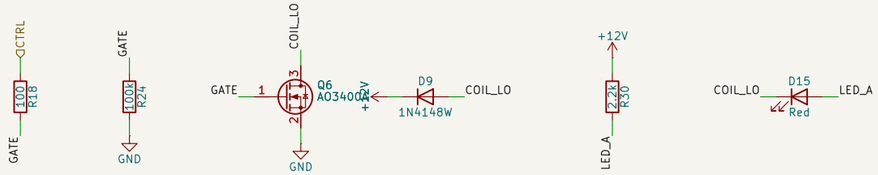


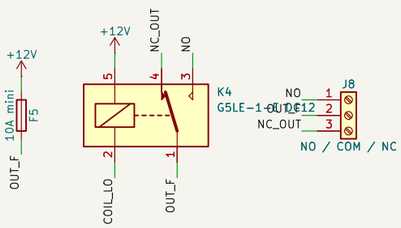


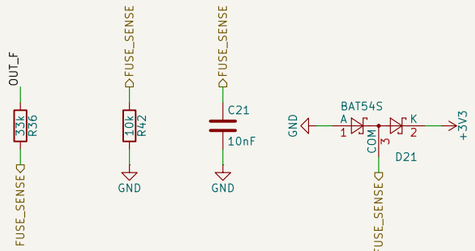


VanController		
Sheet: /Relays/Relay3/		
File: relay_channel.kicad_sch		
Title: Relay channel		
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 7/19

Relay channel. COM is fed from +12V through the channel fuse, so NO gives switched, fused +12 V. FUSE_SENSE is high (~2.8 V at 12 V) while the fuse is intact; BAT54S clamps transients to +3V3.

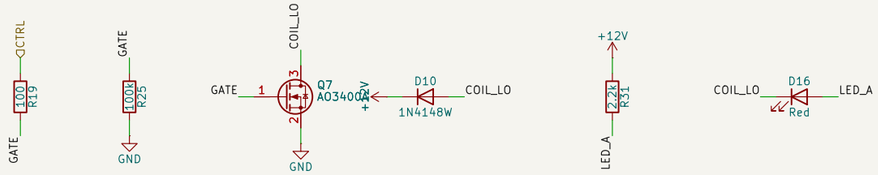


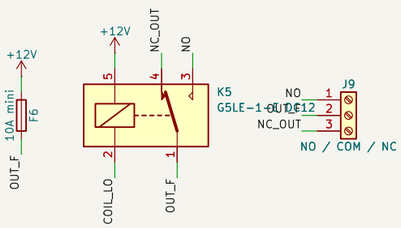


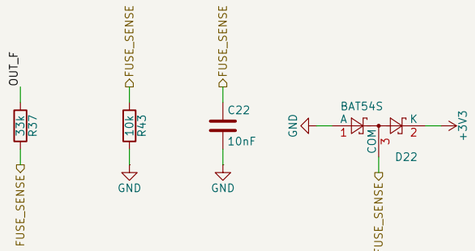


VanController		
Sheet: /Relays/Relay4/		
File: relay_channel.kicad_sch		
Title: Relay channel		
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 8/19

Relay channel. COM is fed from +12V through the channel fuse, so NO gives switched, fused +12 V. FUSE_SENSE is high (~2.8 V at 12 V) while the fuse is intact; BAT54S clamps transients to +3V3.

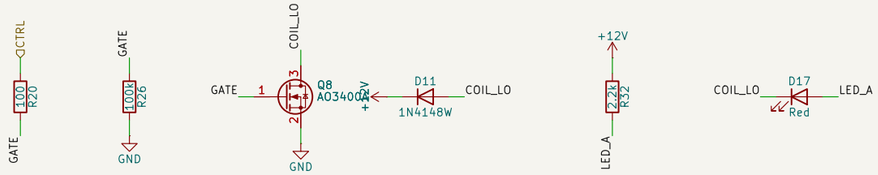


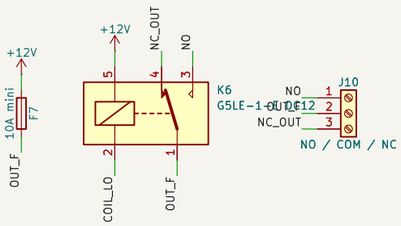


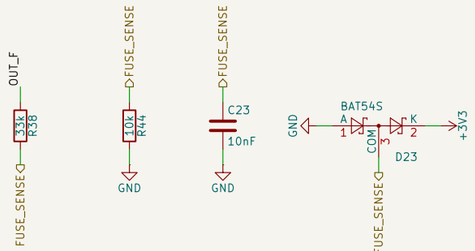


VanController		
Sheet: /Relays/Relay5/		
File: relay_channel.kicad_sch		
Title: Relay channel		
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 9/19

Relay channel. COM is fed from +12V through the channel fuse, so NO gives switched, fused +12 V. FUSE_SENSE is high (~2.8 V at 12 V) while the fuse is intact; BAT54S clamps transients to +3V3.







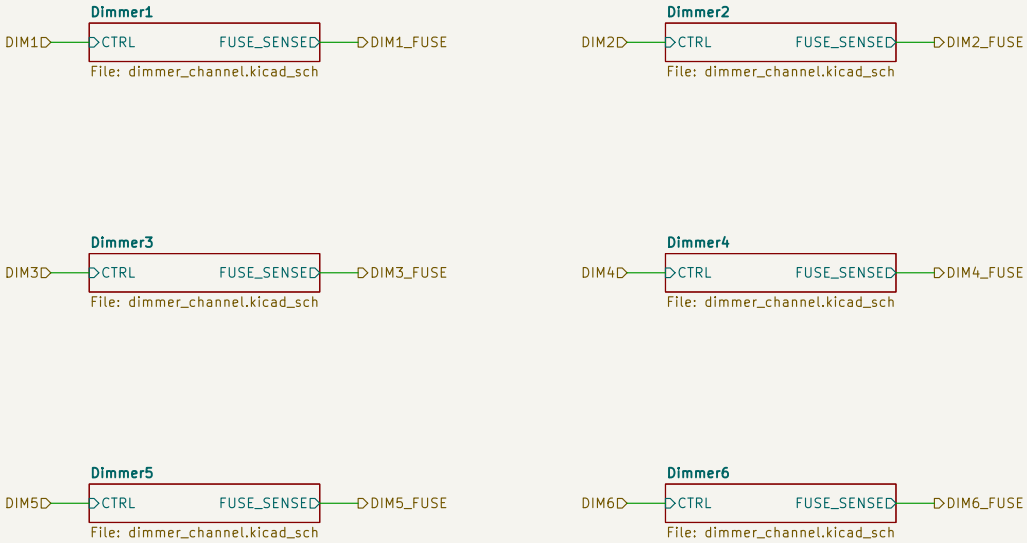
VanController

Sheet: /Relays/Relay6/
File: relay_channel.kicad_sch

Title: Relay channel

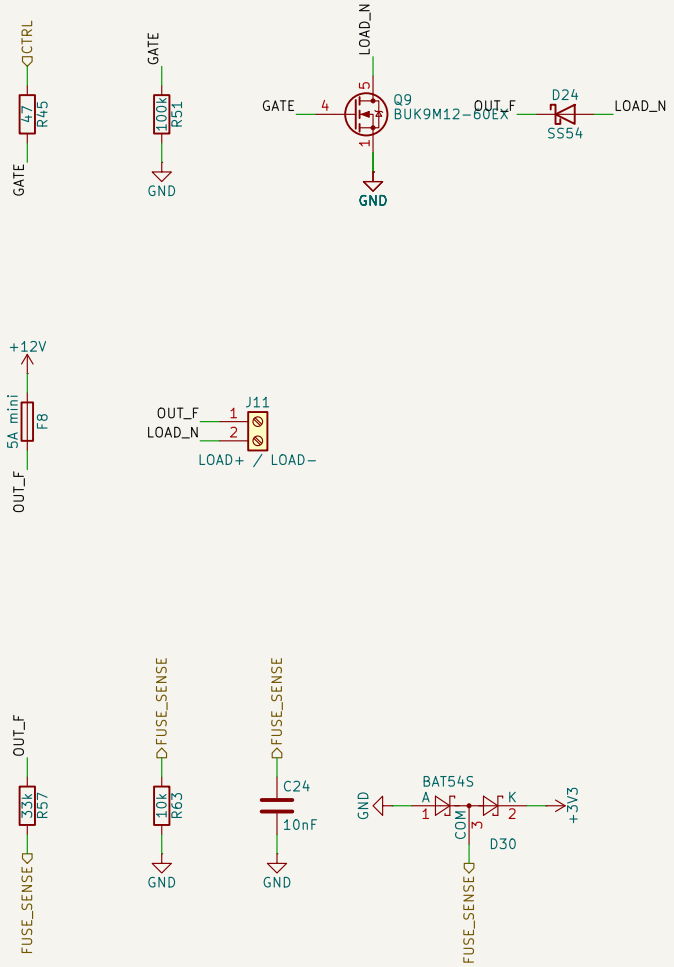
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 10/19

Six identical instances of dimmer_channel.kicad_sch. Edit that file once to change every channel.



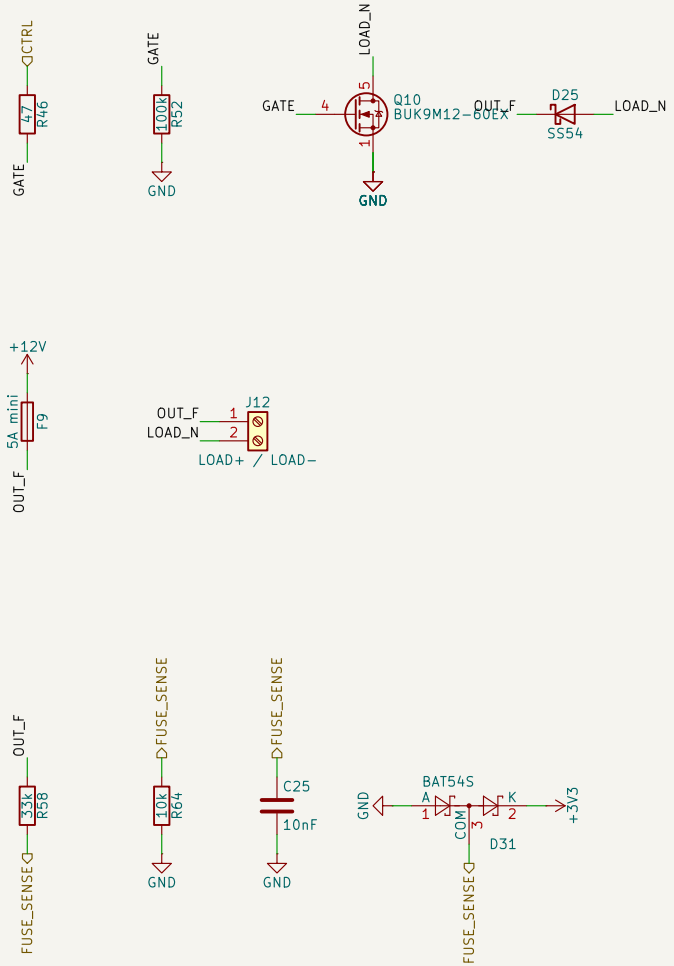
VanController		
Sheet: /Dimmers/ File: dimmers.kicad_sch		
Title: Dimmers (6 channels)		
Size: A4	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 11/19

Dimmer channel, low-side PWM (LEDC, ~20 kHz). LOAD+ is fused +12 V, LOAD- is the MOSFET drain. SS54 freewheels inductive loads.



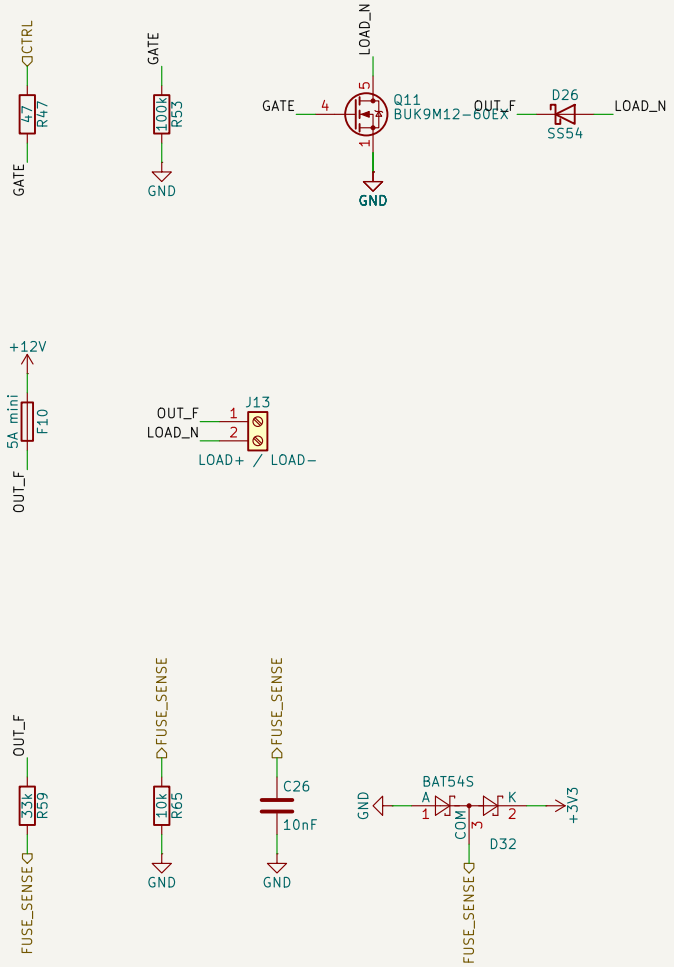
VanController		
Sheet: /Dimmers/Dimmer1/		
File: dimmer_channel.kicad_sch		
Title: Dimmer channel		
Size: A4	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 12/19

Dimmer channel, low-side PWM (LEDC, ~20 kHz). LOAD+ is fused +12 V, LOAD- is the MOSFET drain. SS54 freewheels inductive loads.



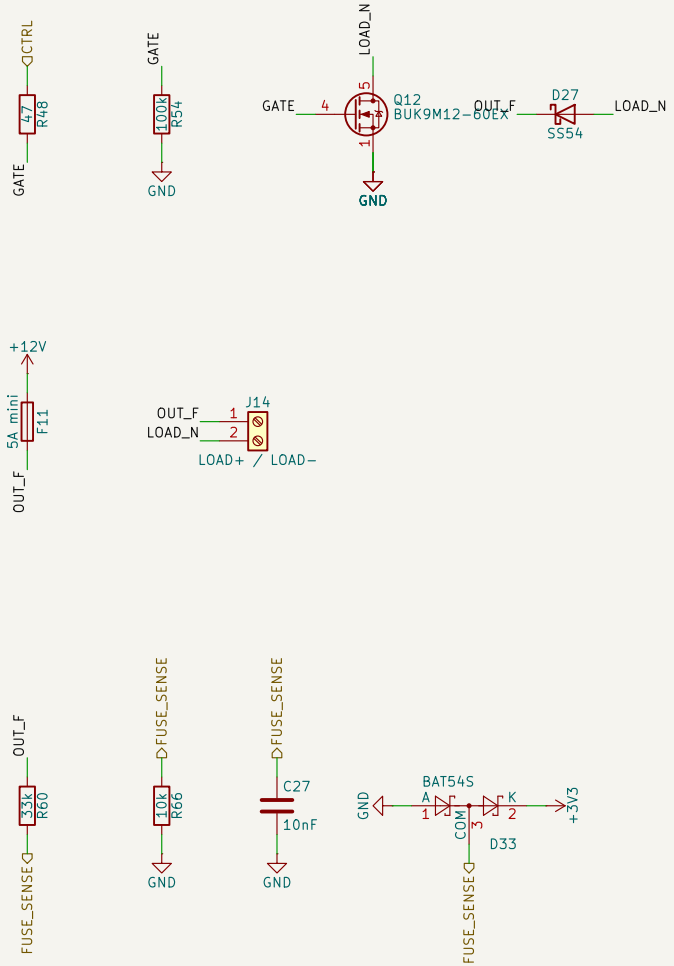
VanController		
Sheet: /Dimmers/Dimmer2/		
File: dimmer_channel.kicad_sch		
Title: Dimmer channel		
Size: A4	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 13/19

Dimmer channel, low-side PWM (LEDC, ~20 kHz). LOAD+ is fused +12 V, LOAD- is the MOSFET drain. SS54 freewheels inductive loads.



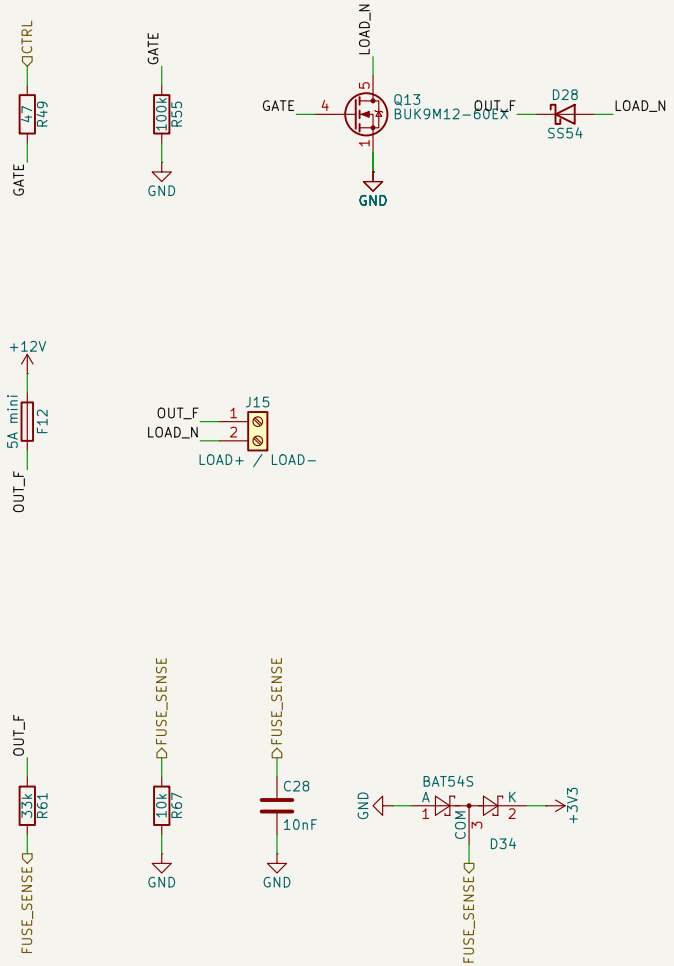
VanController		
Sheet: /Dimmers/Dimmer3/		
File: dimmer_channel.kicad_sch		
Title: Dimmer channel		
Size: A4	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 14/19

Dimmer channel, low-side PWM (LEDC, ~20 kHz). LOAD+ is fused +12 V, LOAD- is the MOSFET drain. SS54 freewheels inductive loads.



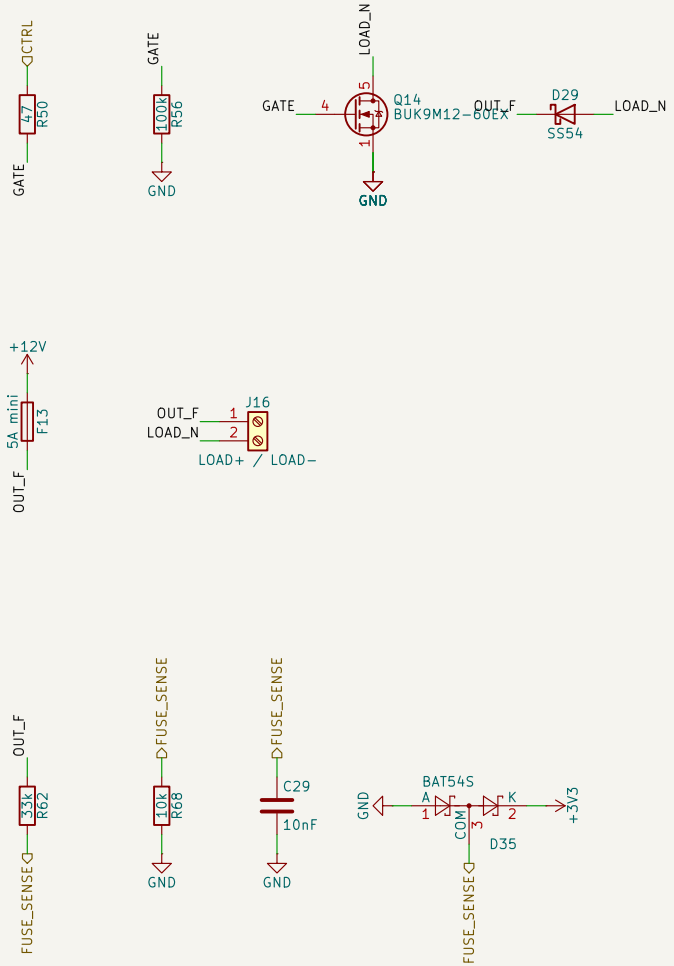
VanController		
Sheet: /Dimmers/Dimmer4/		
File: dimmer_channel.kicad_sch		
Title: Dimmer channel		
Size: A4	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 15/19

Dimmer channel, low-side PWM (LEDC, ~20 kHz). LOAD+ is fused +12 V, LOAD- is the MOSFET drain. SS54 freewheels inductive loads.



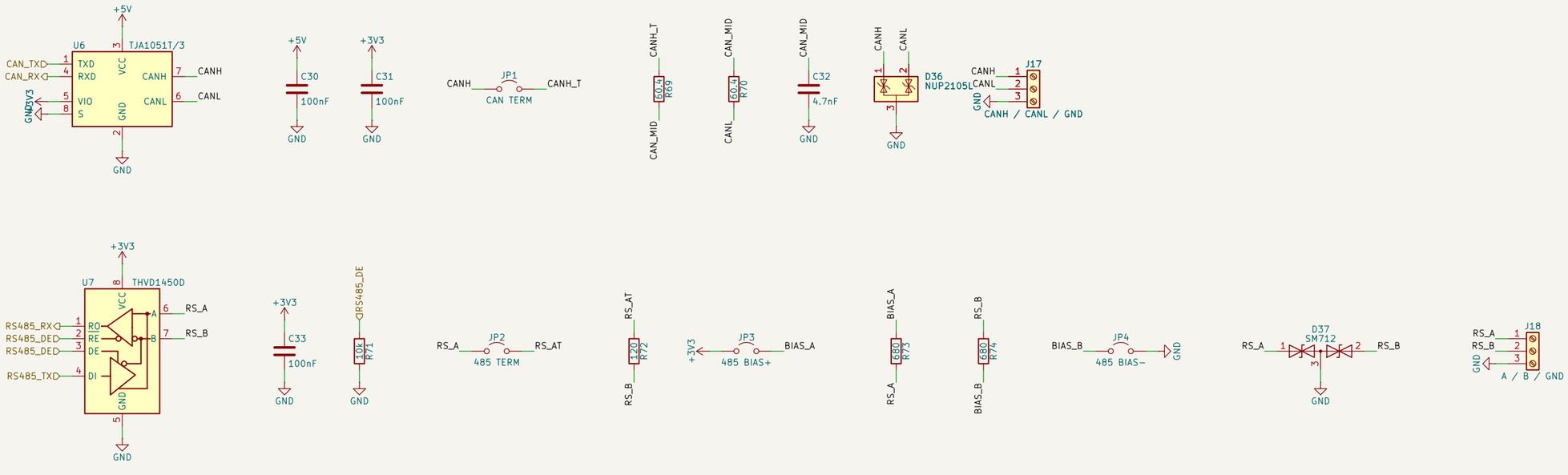
VanController		
Sheet: /Dimmers/Dimmer5/		
File: dimmer_channel.kicad_sch		
Title: Dimmer channel		
Size: A4	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 16/19

Dimmer channel, low-side PWM (LEDC, ~20 kHz). LOAD+ is fused +12 V, LOAD- is the MOSFET drain. SS54 freewheels inductive loads.



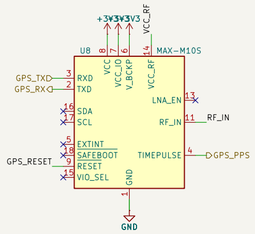
VanController		
Sheet: /Dimmers/Dimmer6/		
File: dimmer_channel.kicad_sch		
Title: Dimmer channel		
Size: A4	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 17/19

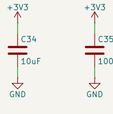
Comms. CAN: TJA1051T/3 (5 V bus side, 3.3 V I/O), split termination on jumper CAN TERM. RS485: THVD1450 at 3.3 V, DE and /RE tied (receive by default), 120 R termination and 680 R fail-safe bias on jumpers.

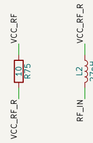


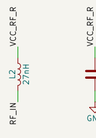
VanController		
Sheet: /Comms/ File: comms.kicad_sch		
Title: Comms		
Size: A3	Date: 2026-10-05	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 18/19

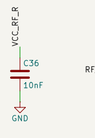
Sensors. Water inputs: fit ONE jumper per input: RESISTIVE (470 R pull-up, e.g. 0–190 R sender), 4–20mA (150 R shunt, loop powered externally), or 0–5V (10k/15k divider). No jumper = direct 0–3.3 V.



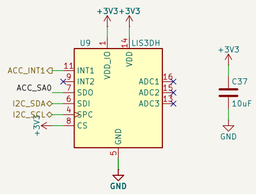


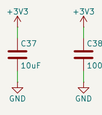




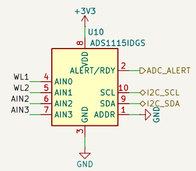


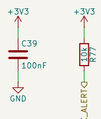














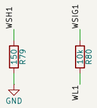






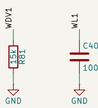




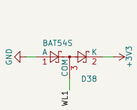








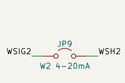


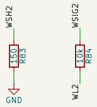




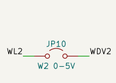


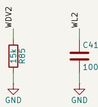




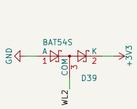


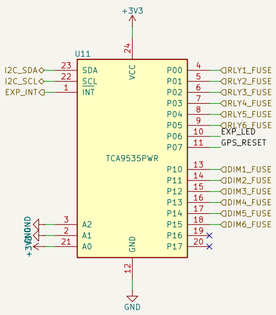




















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